

Volume Segmantics: an annotation-efficient volumetric bioimage segmentation toolkit (version 0.4)

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 03 July 2026

Published: unpublished

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Summary

This is a new version of Volume Segmantics with changes that allow more robust segmentation of complex volumetric bioimages and that make it ready for use in large-scale pipelines for analysis of modalities such as volume electron microscopy (vEM) and cryo-electron tomography (cryo-ET). This release builds on the functionality of the original Volume Segmantics (King et al., 2022), which enabled volumetric segmentation using a multi-axis approach to be usable by biologists and non-machine-learning specialists. This new version enables multi-slice 2.5D segmentation and multi-task training for improved segmentation quality, and the lightweight command-line interface is easy to use in automated and HPC-based workflows. Additionally, enhanced support is provided for an expanded range of encoder and decoder architectures including transformer- and DINO-based encoders, as well as additional loss functions and training options which enable highly configurable pipelines that make efficient use of limited annotations.

Statement of need

Volumetric bio-image segmentation is a key step in the analysis of biological samples imaged with modalities such as 2D and 3D room-temperature or cryogenic transmission electron microscopy (TEM) or scanning electron microscopy (SEM), including approaches such as volume electron microscopy (vEM) and serial section microscopy (McCafferty et al., 2024). Since its release in 2021, Volume Segmantics has fulfilled a need for an open-source tool designed for non-machine-learning specialists to train deep learning models for volumetric image segmentation.

Volume Segmantics now supports the use of multi-slice training and prediction, where a variable number of slices forward and back in the stack are used to train the prediction of the current slice. This additional 3D context often results in smoother segmentations of 3D structures with less undesirable slice-to-slice variation. Secondly, there is support for multi-task training, where models are extended to multi-head and multi-decoder variants, for example to allow training of models on a dataset combining multi-class segmentation, boundary prediction and distance map prediction.

Recent developments in deep learning, including transformer models and foundation models, open opportunities for reducing the amount of annotation required to train a high-performing segmentation model. This new version supports the use of transformer and DINO-based encoders (Caron et al., 2021; Oquab et al., 2023) and integrates that support with its multi-axis, multi-slice and multi-task capabilities. Increasingly, training larger models on larger datasets, as well as the process of hyperparameter optimisation, necessitates the use of HPC.

Overall, Volume Segmantics is a lightweight toolkit built on PyTorch that works well in HPC settings and supports a command-line interface that allows for simple integration into scripted workflows (e.g. for large-scale training).

State of the field

There are a variety of implementations of volumetric segmentation pipelines. Examples include nnU-Net (Isensee et al., 2021), a widely used package that specialises in 3D CNN segmentation; MONAI (Cardoso et al., 2022), a library for medical image segmentation; and micro-SAM (Archit et al., 2025), a tool with GUI support for using Segment Anything (Kirillov et al., 2023) for volumetric images; Empanada, a tool with integrated support for instance segmentation; and Easymode (So-Last et al., 2026), a tool with pretrained models for common EM segmentation tasks. Several commercial packages exist, such as Dragonfly (Comet Technologies Canada Inc., 2025), WebKnossos (Boergens et al., 2017) and Avizo (Thermo Fisher Scientific, 2025), which provide commercial, often proprietary, deep learning frameworks for volumetric image segmentation. Overall, open-source packages often require some coding to use fully, and there are few packages for deep-learning segmentation of volumetric images designed to enable simple command-line creation of configurable and annotation-efficient segmentation pipelines.

New features

In addition to its core functionality of multi-axis prediction using 2D deep learning models, Volume Segmantics has been extended with the following features:

- Multi-slice 2.5D prediction (using an integer number of additional slices).
- Multi-task learning including boundary and distance map / signed distance function (SDF) prediction.
- Confidence map generation using a Test-Time Augmentation method.
- Support for new encoder architectures including Vision Transformer and DINO v1 and v2 architectures (Caron et al., 2021; Oquab et al., 2023).
- Support for loading pretrained encoders and decoders.
- Additional loss functions (class-weighted Dice and boundary loss).
- Support for additional augmentations through MONAI (Cardoso et al., 2022).
- Multi-GPU training and mixed-precision training with PyTorch Lightning.
- Automatic hyperparameter optimisation with Optuna.
- Sliding-window prediction.
- MRC file support and support for Zarr output.
- Improved training metrics, easing the debugging of model training.

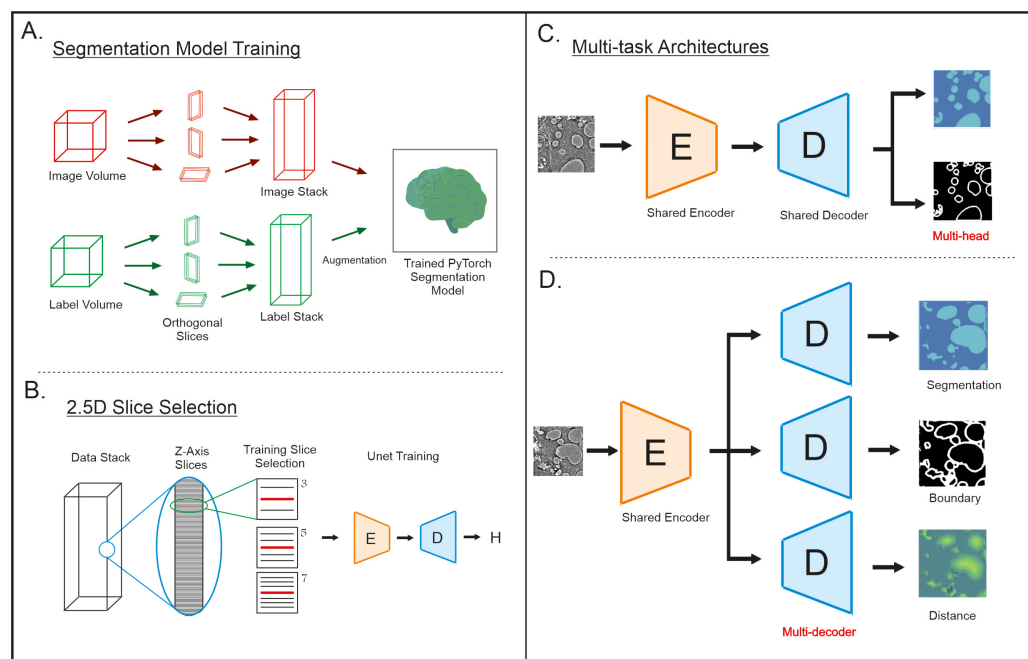


Figure 1: Schematic representation of (A) multi-axis, (B) multi-slice and (C, D) multi-task training supported by Volume Segmantics. The multi-axis training (A) allows improved segmentation performance for isotropic data and supports the confidence map approach described below. The multi-slice training (B) adds additional 3D context that often smooths slice-to-slice noise. Multi-task training (C) involves the prediction of additional targets through, e.g., semantic segmentation and boundary map, and semantic segmentation, boundary map and distance map (D). The additional targets and multi-task training can improve segmentation on particular tasks (e.g. samples with thin boundaries) and the multi-target outputs can flow into post-processing to produce instance segmentations.

Confidence maps

To facilitate error analysis and other downstream geometric evaluation of segmented objects, the new version of Volume Segmantics can provide two extra volumetric outputs: full probability and entropy, that depict the model confidence in a segmentation task. The entropy is normalised based on the number of labels, and the resulting value estimates the uncertainty of an ensemble of test-time averaged predictions (flips and rotations). A higher value for a voxel naturally implies that the trained model is in general less determined about which label to choose.

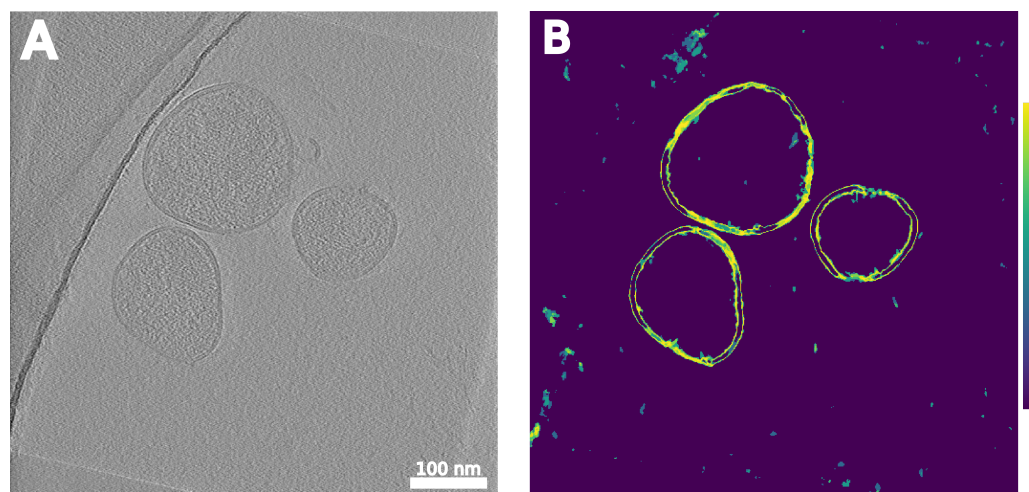


Figure 2: (A) A slice of a 3D cryogenic electron tomogram showing three mature HIV viral particles as an example of the task of segmenting the viral membranes. (B) The entropy map from the prediction of the same slice shown in (A), showing segmentation predictions with high uncertainty (yellow) and low uncertainty (purple). The scale bar is 100 nm and the colour bar ranges from 0 to 1.

Software design

Volume Segmentantics supports configuration via two YAML files where all the training and prediction parameters are set. The software is run by calling two commands, `model-train-2d` and `model-predict-2d`, that accept arguments pointing to training data, specifying training modes and pointing to output locations. Training is monitored using logging in the terminal. When training is finished, several summary plots are saved including an example validation prediction and training loss plots.

The training settings file is divided into several sections, covering the data loading, normalisation, training parameters, loss functions, model architecture and pretrained encoder choice, multi-axis, multi-slice, multi-task, and semi-supervised training options. The prediction file has been extended where necessary to complement the settings in the training file and has sliding-window prediction settings.

In addition, the Optuna configuration settings file supports automated hyperparameter optimisation, with sections covering study settings (name, number of trials, optimisation direction, seed, timeout), pruning, and a search space in which parameters such as model architecture, encoder choice, learning rate schedule, loss function, and training cycle counts are declared with their sampling type and bounds. This can be run by calling `--optuna optuna_config.yaml` after the training argument.

Volume Segmentantics has a Python-based class interface through an API, with a training manager class and a prediction manager class allowing the software to be used as a library in a Jupyter notebook or external application, as in its use within the SuRVoS2 GUI (Pennington et al., 2022).

Implementation and extensibility

Volume Segmentantics is implemented using PyTorch and the MONAI deep learning library and allows augmentations to be performed by either the Albumentations or the MONAI libraries. The software has been driven by Slurm scripts on HPC, where the two commands and respective command-line parameters allow complex segmentation jobs to be scaled up (e.g. training multiple models in a single Slurm script).

Research impact

Volume Segmantics has been used in a variety of applications from cryogenic electron tomography (cryo-ET), to cryo-focused ion beam SEM, to synchrotron X-ray micro-CT (Dumoux et al., 2023; Glynn et al., 2025; Tun et al., 2021). In addition to bio-image segmentation, the Volume Segmantics package can be used in segmentation of other types of volumetric images and has been used recently for concrete aggregate segmentation (Werner et al., 2025) through its interface with SuRVoS2 (Pennington et al., 2022).

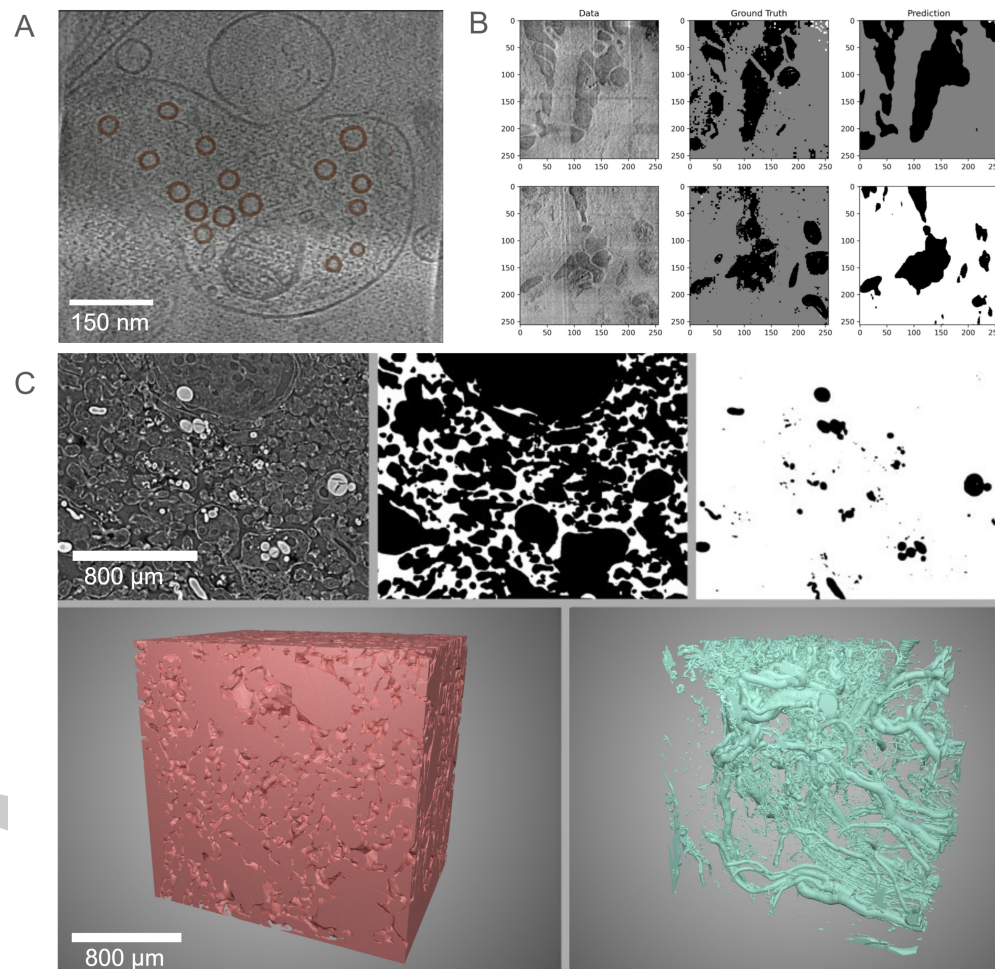


Figure 3: Example of Volume Segmantics use cases and research impact. (A) Cryogenic electron tomography data showing vesicles (orange) from (Glynn et al., 2025), segmented using a ConvNeXtV2 encoder and U-Net architecture. (B) Cryogenic focused ion beam scanning electron microscopy data from (Dumoux et al., 2023) showing segmentation of a challenging imaging modality during development of new imaging approaches. (C) Room-temperature imaging of wax-embedded fixed human placental tissue. Top row, left, shows an image of the reconstructed tomogram from I13 at Diamond Light Source with the labelled fetal vascular network in white pixels, the villous tissue in light grey and the intervillous space in dark grey. Top row, middle and right, villous tissue and fetal vascular segmentations. Bottom row, 3D visualisation of the villous tissue (left, pink) and fetal vascular (right, green) segmentations.

Future developments

There is ongoing work to extend Volume Segmantics to allow additional support for generating segmentation uncertainty maps and support for using additional foundation models. The addition of MONAI for augmentations enables alternative medical and bio-image modality-specific augmentations, and we hope to see more applications of Volume Segmantics to new imaging modalities.

AI usage disclosure

AI code generation (Claude, ChatGPT, Co-pilot) was used at several points of the development process, though all code was reviewed by and all core software design decisions were made by the authors. All segmentation method validation and scientific interpretation was performed and reviewed by the authors. Claude was used for formatting the paper and bibliography. All generative AI output was critically reviewed and manually revised by the authors.

Acknowledgements

The authors acknowledge support from BBSRC grant number BB/Y005953/1 (MCD) and the Wellcome Leap *In Utero* program (MCD). Synchrotron CT imaging of human placenta was performed on Beamline I13-2 of the Diamond Light Source synchrotron in Oxfordshire, UK. The testing of the software and some of the segmentations were performed using the Baskerville Tier 2 HPC service (<https://www.baskerville.ac.uk/>). Baskerville was funded by the EPSRC and UKRI through the World Class Labs scheme (EP/T022221/1) and the Digital Research Infrastructure project (EP/W032244/1) and is operated by Advanced Research Computing at the University of Birmingham.

The authors would like to thank Maud Dumoux, Audrey Le Bas, and Ana Katsini for support in testing the software, as well as Rohan Lewis, Davis Laundon, Bram Sengers and Aaron Grewal for enlightening discussions about how to improve segmentation of placental villi and villous trees.

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